REGIONALIZATION

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WHY POOL STATIONS

- Nearby stations should usually have similar rainfall probabilities.
  - The same holds for floods, fire, and other hazards.
- Pooling reduces some forms of sampling error:
  - where a severe thunderstorm went
  - where a hurricane's rain bands persisted
- Some sampling variability remains:
  - there's one Harvey in the region

BIAS-VARIANCE TRADE-OFF

- One station's record gives a nearly **unbiased** but noisy (high-variance) estimate.
- **Regionalization** makes some or all parameters similar or identical across the stations in a region.
- Fewer stations: less bias, more variance.
- More stations: less variance, possibly more bias.

KEY CONSIDERATIONS FOR REGIONALIZATION

1. What data do I have?
2. **Physically**, what is driving sampling variability and can I reduce it with pooling?
3. What are “similar” stations?
4. Which parameters should be pooled?

L-MOMENTS, RECALLED

- Linear combinations of order statistics.
- Fit by choosing the parameters so that the distribution's L-moments match the data's (Hosking, 1990).
- Pros:
  - Fast: a closed form, no optimizer.
  - Robust: they work well in practice, even on short records.
- Cons:
  - Rigid: hard to add covariates or priors.
  - Hard to attach uncertainty to the parameters.

REGIONAL FREQUENCY ANALYSIS (INDEX FLOOD)

Idea: every station in the region has the same shape of **flood** distribution, up to a scaling factor (Hosking, 1997).

1. Divide each station's annual maxima by its mean, m_s (the *index flood*).
2. Average the stations' L-CV and L-skewness, weighted by record length, and fit one GEV to them: the regional growth curve $q(T)$.
3. Scale back to each station: $x_T = m_s q(T)$.

Fails when stations differ in more than size: for U.S. streamflow, variability falls as catchment area grows (Hosking, 1997).

REGION OF INFLUENCE

Goal: the best estimate at one target station, using regional information (Hosking, 1997).

1. Include the stations that are reasonably similar to the target: nearby, at a similar elevation, in a similar climate.
2. Weight each station j by its similarity to the target, w_j.
3. Fit by maximizing the weighted log-likelihood:

$$\ell(\theta) = \sum_j w_j \sum_t \log f(y_{j,t} \mid \theta)$$

A common two-step version fixes the shape at its regional estimate, then fits the station's location and scale from its own record.

WHICH STATIONS ARE SIMILAR?

Ask

A station's skewness is 0.10; its three nearest neighbors' are 0.37, 0.07, and 0.27. Should all three be in its region?

HOW ATLAS 14 AND ATLAS 15 CHOOSE STATIONS

i Note

Figure shown in class: Atlas 15's weighted region for one target station (black square): black stations get full weight, gray partial, and white zero, and some white stations sit close to the target, from the NOAA Atlas 15 pilot report, Fig. 6 (Perica et al., 2024).

- Atlas 14: the 15 nearest stations, edited by an analyst for ridges, elevation, and mean annual maximum (Perica et al., 2018).
- Atlas 15, a preliminary report, weights every station within 160 km (100 miles) by distance, rainfall climate, elevation, and terrain between (Perica et al., 2024).